

3) We should consider

$$|u(x, t) - u_0(x_0)| \rightarrow 0 \text{ as } (x, t) \rightarrow (x_0, 0)$$

$$|u(x, t) - u_0(x_0)|$$

$$\stackrel{(12.5)}{=} \left| \int_{\mathbb{R}^n} \frac{1}{(4\pi t)^{n/2}} \exp\left(-\frac{|y-x|^2}{4t}\right) u_0(y) dy - u_0(x_0) \right|$$

Note that

$$\int_{\mathbb{R}^n} \frac{1}{(4\pi t)^{n/2}} \exp\left(-\frac{|x|^2}{4t}\right) dx = 1.$$

for any  $t > 0$  we have

$$|u(x, t) - u_0(x_0)|$$

$$\begin{aligned} &= \left| \int_{\mathbb{R}^n} \frac{1}{(4\pi t)^{n/2}} \exp\left(-\frac{|y-x|^2}{4t}\right) u_0(y) dy \right. \\ &\quad \left. - \int_{\mathbb{R}^n} \frac{1}{(4\pi t)^{n/2}} \exp\left(-\frac{|y-x|^2}{4t}\right) u_0(x_0) dy \right| \\ &= \left| \int_{\mathbb{R}^n} \frac{1}{(4\pi t)^{n/2}} \exp\left(-\frac{|x-y|^2}{4t}\right) (u_0(y) - u_0(x_0)) dy \right| \\ &\leq \left| \int_{B_\delta(x_0)} \frac{1}{(4\pi t)^{n/2}} \exp\left(-\frac{|x-y|^2}{4t}\right) (u_0(y) - u_0(x_0)) dy \right| \\ &\quad + \left| \int_{\mathbb{R}^n \setminus B_\delta(x_0)} \frac{1}{(4\pi t)^{n/2}} \exp\left(-\frac{|x-y|^2}{4t}\right) (u_0(y) - u_0(x_0)) dy \right| \end{aligned}$$

*small* *small*

$$=: I_\delta + J_\delta \quad (\delta > 0).$$

$$I_\delta: |y - x_0| < \delta \Rightarrow u_0(y) - u_0(x_0) : \text{small} \Rightarrow I_\delta \rightarrow 0.$$

$$J_\delta: |y - x_0| < \delta$$

$$\Rightarrow \exp\left(-\frac{|y-x_0|^2}{4t}\right) : \text{small} \Rightarrow J_\delta \rightarrow 0.$$

## Chapter 13 ~~hom~~ Inhomogeneous heat equations and Duhamel's principle

Duhamel's formula for ODEs.

$$(13.1) \quad \begin{cases} v'(t) - av(t) = 0, & t > 0 \\ v(0) = g \end{cases}$$

$$v: [0, \infty) \rightarrow \mathbb{R} \text{ unknown}$$

$$g, a \in \mathbb{R} \text{ given.}$$

What is a sol of (13.1)?

Multiply  $e^{-at}$

$$\xrightarrow{\quad} e^{-at} v'(t) - \underbrace{ae^{-at}}_{+(e^{-at})'} v(t) = 0.$$

$$\rightarrow (e^{-at} v(t))' = 0.$$

$$\therefore e^{-at} v(t) = e^{-a \cdot 0} v(0) \quad \forall t > 0.$$

$$= v(0) = g \quad (13.1)$$

Ans.  $v(t) = e^{at} g$  is a sol of (13.1)

Here we define.

$$S_t: \mathbb{R} \rightarrow \mathbb{R}$$

$$\Downarrow$$

$$g \mapsto e^{at} g \quad (t > 0)$$

$S_t$  is sometimes called "solution map".

Next we consider

$$(13.2) \quad \begin{cases} w'(t) - aw(t) = f(t), & t > 0 \\ w(0) = 0 \end{cases}$$

$$w: [0, \infty) \rightarrow \mathbb{R} \text{ unknown}$$

$$f: [0, \infty) \rightarrow \mathbb{R} \text{ given.}$$

What is a sol of (13.2)?

Multiply  $e^{-at}$

$$e^{-at} w'(t) - \underbrace{a \cdot e^{-at} w(t)}_{+(e^{-at})'} = e^{-at} f(t)$$

$$\rightarrow e^{-at} w'(t) + (e^{-at})' w(t) = e^{-at} f(t)$$

$$\rightarrow (e^{-at} w(t))' = e^{-at} f(t)$$

change  $t \leftrightarrow s$

$$(e^{-as} w(s))' = e^{-as} f(s)$$

Integrate over  $0 \leq s \leq t$

$$\int_0^t (e^{-as} w(s))' ds = \int_0^t e^{-as} f(s) ds$$

||

$$\left[ e^{-as} w(s) \right]_0^t$$

$$\stackrel{||}{=} e^{-at} w(t) - \underbrace{e^{-a \cdot 0} w(0)}_{=0 \text{ (}\because (13.2)\text{)}}$$

Ans:  $w(t) = e^{at} \int_0^t e^{-as} f(s) ds$  is a sol of (13.2)

$$\Rightarrow w(t) = \int_0^t e^{a(t-s)} f(s) ds$$

$$= \int_0^t S_{t-s} f(s) ds.$$

This means that sol of (13.2) can be obtained by "sol map" of (13.1). This method is called Duhamel's principle. By the linearity of the problem.

$$(13.3) \begin{cases} u'(t) - au(t) = f(t), & t > 0 \\ u(0) = g \end{cases}$$

has a sol.

$$u(t) = v(t) + w(t) = S_t g + \int_0^t S_{t-s} f(s) ds.$$

Remark:  $S_t : \mathbb{R} \rightarrow \mathbb{R}$

$$\begin{matrix} u \\ g \end{matrix} \mapsto \begin{matrix} u \\ e^{at} g \end{matrix}$$

We often write  $S_t$  as  $e^{at}$ :  $\underset{\text{map}}{S_t} = \underset{\text{scalar}}{e^{at}}$

<Inhomogeneous heat eq>

Consider

$$(*) \begin{cases} \frac{\partial w}{\partial t}(x, t) - \Delta w(x, t) = f(x, t), & x \in \mathbb{R}^n, t > 0 \\ w(x, 0) = 0 \end{cases}$$

Recall the last lecture that.

$$\begin{cases} \frac{\partial v}{\partial t}(x, t) - \Delta v(x, t) = 0, & x \in \mathbb{R}^n, t > 0 \\ v(x, 0) = g(x) \end{cases} \quad \leftarrow (13.1), (13.2) \text{に形が似ている。}$$

has a sol of the form

$$v(x, t) = \int_{\mathbb{R}^n} \Phi(x-y, t) g(y) dy$$

Here we denote

$$e^{t\Delta} : g \mapsto \int_{\mathbb{R}^n} \Phi(x-y, t) g(y) dy$$

That is

$$(e^{t\Delta})(x) := \int_{\mathbb{R}^n} \Phi(x-y, t) g(y) dy$$

Then, from Duhamel formula for ODE.

one can expect

$$w(x, t) = \int_0^t e^{(t-s)\Delta} f(0, s) ds \text{ is a sol of (*)}$$

$\Rightarrow$  Ans: true if  $f$  is smooth and has a cpt. support.

Here

$$e^{(t-s)\Delta} f(0, s) = \int_{\mathbb{R}^n} \Phi(x-y, t-s) f(y, s) dy \quad (13.4)$$

hence

$$\begin{aligned} w(x, t) &= \int_0^t e^{(t-s)\Delta} f(0, s) ds \\ &= \int_0^t \left( \int_{\mathbb{R}^n} \Phi(x-y, t-s) f(y, s) dy \right) ds \\ &= \int_0^t \left( \int_{\mathbb{R}^n} \frac{1}{(4\pi(t-s))^{n/2}} \exp\left(-\frac{(x-y)^2}{4(t-s)}\right) f(y, s) dy \right) ds \end{aligned}$$

Thm 13.1

Let  $f = f(x, t) : \mathbb{R}^n \times [0, \infty) \rightarrow \mathbb{R}$  be a given func. such that  $f$  is  $C^2$  with respect  $x$ ,  $C^1$  with respect to  $t$ , and  $f$  has cpt. support. Let  $w$  be defined by (13.4)

Then the following holds

1)  $w$  is  $C^2$  w.r.t.  $x$   
 $C^1$  w.r.t.  $t$

2)  $\frac{\partial w}{\partial t}(x, t) - \Delta w(x, t) = f(x, t)$ ,  $x \in \mathbb{R}^n, t > 0$ .

3) For fixed  $x_0 \in \mathbb{R}^n$

$w(x, t) \rightarrow 0$  as  $(x, t) \rightarrow (x_0, 0)$   $t > 0$ .

pf. Consider 2). only

1.) ~~Compute~~ Compute the derivative of  $w$

Take a change of variable

$$y \leftarrow x - y, \quad s \leftarrow t - s$$

We have

$$w(x, t) = \int_0^t \left( \int_{\mathbb{R}^n} \frac{1}{(4\pi s)^{n/2}} \exp\left(-\frac{|y|^2}{4s}\right) f(x-y, t-s) dy \right) ds$$

Therefore

$$\begin{aligned} \frac{\partial^2 w}{\partial x_i \partial x_j}(x, t) &= \int_0^t \left( \int_{\mathbb{R}^n} \frac{1}{(4\pi s)^{n/2}} \exp\left(-\frac{|y|^2}{4s}\right) \frac{\partial^2 f}{\partial t}(x-y, t-s) dy \right) ds \\ &\quad + \int_{\mathbb{R}^n} \frac{4}{(4\pi s)^{n/2}} \exp\left(-\frac{|y|^2}{4t}\right) f(x-y, 0) dy \end{aligned}$$

2) Compute  $\frac{\partial w}{\partial t} - \Delta w$

$$\begin{aligned} &\frac{\partial w}{\partial t}(x, t) - \Delta w \\ &= \left( \int_0^\varepsilon + \int_\varepsilon^t \right) \left( \int_{\mathbb{R}^n} \frac{1}{(4\pi s)^{n/2}} \exp\left(-\frac{|y|^2}{4s}\right) \left( \frac{\partial f}{\partial t}(x-y, t-s) - \Delta f(x-y, t-s) \right) dy \right) ds \\ &\quad + \int_{\mathbb{R}^n} \frac{1}{(4\pi t)^{n/2}} \exp\left(-\frac{|y|^2}{4t}\right) f(x-y, 0) dy \\ &=: J_\varepsilon + I_\varepsilon + K \quad \text{for } \varepsilon > 0. \end{aligned}$$

3) Estimate  $J_\varepsilon$

$$|J_\varepsilon| \leq \int_0^\varepsilon \left( \int_{\mathbb{R}^n} \frac{1}{(4\pi s)^{n/2}} \exp\left(-\frac{|y|^2}{4s}\right) dy \right) ds$$

$$\sup_{\substack{x \in \mathbb{R}^n \\ t > 0}} \left| \frac{\partial f}{\partial t}(x, t) - \Delta f(x, t) \right|$$

$=: C$

$$= C_\varepsilon \rightarrow 0 \quad (\varepsilon \downarrow 0).$$

4) Compute  $I_\varepsilon$ :

Note that

$$\begin{aligned} \frac{\partial f}{\partial t}(x-y, t-s) &= \frac{\partial}{\partial t} f(x-y, t-s) \\ &= -\frac{\partial}{\partial s} f(x-y, t-s) \end{aligned}$$

$$\begin{aligned} \Delta f(x-y, t-s) &= \sum_i \frac{\partial}{\partial x_i} \cdot \frac{\partial}{\partial x_i} f(x-y, t-s) \\ &= \sum_i \frac{\partial}{\partial y_i} \cdot \frac{\partial}{\partial y_i} f(x-y, t-s) \\ &= \Delta_y (f(x-y, t-s)) \end{aligned}$$

Then by the integration by parts.

$$\begin{aligned} &\int_\varepsilon^t \int_{\mathbb{R}^n} \frac{1}{(4\pi s)^{n/2}} \exp\left(-\frac{|y|^2}{4s}\right) \left( \frac{\partial f}{\partial t}(x-y, t-s) - \Delta f(x-y, t-s) \right) dy ds \\ &= \int_\varepsilon^t \int_{\mathbb{R}^n} \left( \frac{\partial}{\partial s} - \Delta_y \right) \left( \frac{1}{(4\pi s)^{n/2}} \exp\left(-\frac{|y|^2}{4s}\right) \right) f(x-y, t-s) dy ds \\ &\quad - \int_{\mathbb{R}^n} \frac{1}{(4\pi t)^{n/2}} \exp\left(-\frac{|y|^2}{4t}\right) f(x-y, 0) dy \\ &\quad + \int_{\mathbb{R}^n} \frac{1}{(4\pi \varepsilon)^{n/2}} \exp\left(-\frac{|y|^2}{4\varepsilon}\right) f(x-y, t-\varepsilon) dy \end{aligned}$$

$= K$

About Assingment?

1. mail (前日までに)

2. 対面 (-12:20)

Thus

$$\begin{aligned} I_\varepsilon &= -K + \int_{\mathbb{R}^n} \frac{1}{(4\pi \varepsilon)^{n/2}} \exp\left(-\frac{|y|^2}{4\varepsilon}\right) \cdot f(x-y, t-\varepsilon) dy \\ &\rightarrow -K + f(x, t) \quad (\varepsilon \downarrow 0) \quad \text{by Thm 12.2} \end{aligned}$$

$$\therefore I_\varepsilon + J_\varepsilon + K \rightarrow -K + f(x, t) + K = f(x, t) \quad \square$$

Chapter 14: Maximum principle for heat eq

Thm 14.1

Let  $\Omega \subset \mathbb{R}^n$  be a bdd open set.

Let  $u: \overline{\Omega} \times [0, T] \rightarrow \mathbb{R}$  be a conti func. up to b'dry (conti on  $\overline{\Omega} \times [0, T]$ )

such that  $u$  is twice diff'ble w.r.t  $x$  and is a diff'ble w.r.t.

If  $u$  satisfies

$$\frac{\partial u}{\partial t}(x, t) - \Delta u(x, t) = 0, \quad x \in \Omega, t > 0. \quad (14.1)$$

then

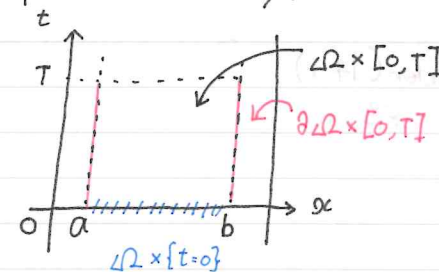
$$\sup_{(x,t) \in \Omega \times [0,T]} u(x,t) = \sup_{(x,t) \in P_B(\Omega \times [0,T])} u(x,t). \quad (14.2)$$

where

$$P_B(\Omega \times [0, T]) := (\partial\Omega \times [0, T]) \cup (\Omega \times \{t=0\})$$

is called "parabolic boundary".

< parabolic boundary >



$$\Omega = (a, b) \in \mathbb{R}$$